

Wednesday, 8 July, 2026 at 3 pm CEST / 1 pm UTC

Registration Link: <https://washington.zoom.us/meeting/register/MbV5ougcSqWOJBd0NF-oHw>

Meeting ID: 976 9119 4859 • Passcode: 634738

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SCOR Working Group 168 Webinar #10

Title: Neural-BGC: Neural Network emulation of BGC from T/S data for interpolation and coupled Physics-BGC simulations

Presenter:

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Ocean biogeochemistry (BGC) regulates key variables such as oxygen and nutrients that are essential for marine ecosystems and the global carbon cycle. However, BGC observations remain sparse, and BGC models rely on poorly constrained parameters and processes. To address these limitations, we developed **Neural-BGC**, an AI model trained on millions of ocean observations collected since 1965. Neural-BGC emulates BGC variables from temperature and salinity fields. It is used both for interpolation and coupled physics-BGC simulations. For interpolation, the model estimates BGC variables from available temperature and salinity observations, and these estimates are combined with direct oxygen measurements through optimal interpolation to produce gridded products. When coupled to a physical ocean model, Neural-BGC reproduces biogeochemical features, including the oxygen minimum zones of the Indian Ocean and nutrient-rich blooms in the Canary Current Upwelling System.

Webinar Series Information

The [4D-BGC Working Group](#) seeks to enhance access and utility of Biogeochemical (BGC) Argo observations through four-dimensional (4D) data products. These advanced data products aim to refine our understanding of ocean biogeochemistry, improve biogeochemical models and reanalysis products, and provide valuable insights for policy-making. The goal of this webinar series is to introduce new and in-development BGC data products, review techniques used to develop data products from in situ observations, and to explore ways in which 4D-BGC products are leveraged to answer scientific questions.